based on predefined policy rules. Traffic shaping uses traffic classification, policy rules, queue disciplines and quality of service (QoS).

The need for traffic shaping arises because network bandwidth is an expensive resource that is shared among many parties in an organisation, and some applications require guaranteed bandwidth and priority. Traffic shaping lets you: (1) control network services, (2) limit bandwidths, and (3) guarantee Quality of Service (QoS). Intelligently managed traffic shaping improves network latency, service availability and bandwidth utilisation.

NetEM capabilities include delay (it delays each packet); loss (it drops some packets); duplication (it duplicates some packets); and corruption (it introduces a single bit error at a random offset in a packet).

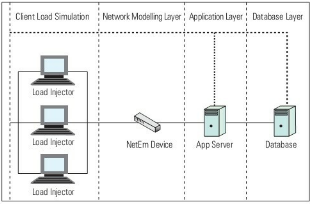

Figure 1: Current environment

How NetEM works

NetEM consists of two components—a tiny kernel module for a queuing discipline, and a command-line utility to configure it. The kernel module has been integrated in 2.6.8 (and 2.4.28) or later, and the command is part of the iproute2 package. The command-line utility communicates with the kernel via the netlink socket interface. It encodes its requests into a standard message format, which the kernel decodes.

The queuing layer exists between the network device and the protocol output. The default queuing discipline is a simple FIFO packet queue. Queuing discipline consists of two key interfaces; one queues packets to be sent, and the other releases packets to the network device for transmission. The queuing discipline makes the policy decision of which packets to send, based on the current settings.

Terminology

qdisc	A queue discipline (qdisc) is a set of rules that determine the order in which arrivals are serviced. It is a packet queue with an algorithm that decides when to send each packet.
Classless qdisc	A qdisc with no configurable internal sub-division.
Classful qdisc	A qdisc that may contain classes; classful qdiscs allow packet classification (Class-Based Queueing and others)
Root qdisc	The root qdisc is attached to each network interface—either classful or classless.
egress qdisc	Works on outgoing traffic only.
ingress qdisc	Works on incoming traffic.
Filter	Classification can be performed using filters

Here are some examples that can be used for network throttling.

Network emulation examples

Note: I have used *eth1* for the interface in the examples below; you should use the name of the specific Ethernet card where traffic needs to be throttled.

To add *constant delay* to every packet going out through a specific interface, use the following command:

```
# tc qdisc add dev eth1 root netem delay 80ms
```

Now a ping test to this host should show an increase of 80ms in the delay to replies.

To add *random variance*, use the command below:

```
# tc qdisc change dev eth1 root netem delay 80ms 10ms
```

We can also add *variable delay (jitter)/Random Variance* too. Most wide-area networks like the Internet have some jitter associated with them. The following command will add +/- 10 ms of jitter to the 80 ms of delay.

```
# tc qdisc add dev eth1 root netem delay 80ms 10ms
```

To *see* what queueing discipline (qdisc) has been applied to an interface, use:

```
# tc qdisc show dev eth1
```

To *turn off/delete* the qdisc from a specific interface (in this case, *eth1*), execute the command given below:

```
# tc qdisc del dev eth1 root
```

Typically, the delay in a network is not uniform. It is more common to use something like a normal distribution to describe the variation in delay. NetEM can accept a non-uniform distribution:

```
# tc qdisc change dev eth1 root netem delay \
100ms 20ms distribution normal
```